

0 2

A cyclist travels at constant speed round a circular track of radius 42 m.
She completes one circuit of the track every 24 s.

0 2 . 1

Calculate the speed of the cyclist.

[2 marks]

speed = _____ m s^{-1}

0 2 . 2

Deduce the magnitude of her average velocity over a 24 s period while travelling once round the track.

[1 mark]

magnitude of average velocity = _____ m s^{-1}

0 2 . 3

Calculate the magnitude of her average velocity over a 12 s period while she travels halfway round the circular track.

[2 marks]

magnitude of average velocity = _____ m s^{-1}



0 3 . 1 Complete **Table 1** to show some of the properties of an antiproton.

[2 marks]

Table 1

Charge / C	Rest mass / kg	Specific charge / C kg⁻¹

0 3 . 2 Antiprotons can be annihilated.

Describe annihilation.
Include the circumstances under which it can occur and its outcome.

[4 marks]



Do not write

2

4

In the Rutherford scattering experiment, most of the alpha particles

[1 mark]

A were deflected by an angle of more than 90° .

☐

B were attracted to the gold nuclei.

☐

C were stopped by the gold foil.

☐

D passed straight through the gold foil undeflected.

☐

Do not write
outside the
box

Complete **Table 1** with the charge of each particle and the name of its corresponding antiparticle.

[2 marks]

Table 1

Particle	Charge / C	Antiparticle
Positron		
Neutron		

A proton and an antiproton are travelling in opposite directions with the same speed. They collide and annihilate.

State and explain the characteristics of the radiation produced in the annihilation.

[4 marks]



Do not write

0 7 A nucleus of an isotope of radium decays to produce a β^- particle, a nucleus of actinium and a particle **X**.

0 7 . 1 Identify **X**.

[1 mark]

0 7 . 2 State **two** characteristics of **X** that make it more difficult to detect than the β^- particle.

[2 marks]

1

2

0 7 . 3 Explain why the specific charge on the actinium nucleus is greater than the specific charge on the radium nucleus.

[2 marks]



0 7 . 4

The nucleus of a different isotope of radium has a specific charge of $3.784 \times 10^7 \text{ C kg}^{-1}$.

Assume that the mass of a nucleon in the nucleus is $1.661 \times 10^{-27} \text{ kg}$.

Determine the number of neutrons in the radium nucleus.

proton number of radium = 88

[3 marks]

number of neutrons = _____

8

Turn over for the next question

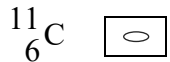
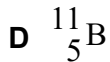
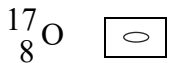
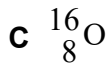
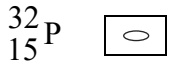
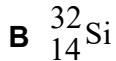
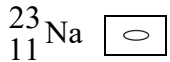
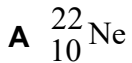
Turn over ►



Do not write

Which pair of nuclei has the same number of neutrons?

[1 mark]



2 0 Which is a possible annihilation event?

[1 mark]

A $p + \bar{p} \rightarrow \gamma$



B $p + p \rightarrow \gamma + \gamma$



C $e^{-} + e^{+} \rightarrow \gamma$



D $e^{-} + e^{+} \rightarrow \gamma + \gamma$



Calculate the specific charge of an alpha particle.

[2 marks]

specific charge = _____ C kg⁻¹



Don't write

The Rutherford scattering experiment provided evidence for a new model of the atom.

What is different about Rutherford's model compared with the previous model?

[1 mark]

A The atom is neutrally charged overall.

☐

B The atom contains negatively charged electrons.

☐

C The negative charge is concentrated in the centre of the atom.

☐

D The mass is concentrated in the centre of the atom.

☐

2 1

An anti-hydrogen atom consists of a positron and an anti-proton.

Which row shows the charge and the mass of the anti-hydrogen atom?

[1 mark]

	Charge / C	Mass / kg	
A	zero	-1.7×10^{-27}	☐
B	zero	1.7×10^{-27}	☐
C	3.2×10^{-19}	1.7×10^{-27}	☐
D	3.2×10^{-19}	-1.7×10^{-27}	☐

2**2**

Which quantity is conserved during pair production?

[1 mark]

A kinetic energy

☐

B mass

☐

C momentum

☐

D velocity

☐

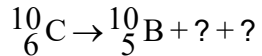
Turn over ►



Do not write Do not write

2**3**

What are the missing particles in this decay equation?



[1 mark]

A electron and neutrino

B electron and antineutrino

C positron and neutrino

D positron and antineutrino

0 1

On one occasion, an electron and a positron are created by pair production when a high-energy photon interacts with the nucleus of an atom.

The rest energy of an electron is 0.51 MeV.

0 1 . 1

Show that the photon must have a minimum energy of approximately 1.6×10^{-13} J.

[1 mark]

0 1 . 2

On another occasion, an electron–positron pair is created by a photon with an energy of 2.1×10^{-13} J.

Deduce how energy is conserved on this occasion.

[2 marks]

0

2

1

Explain why momentum is a vector quantity.

[1 mark]

*Do not write
outside the
box*

One conclusion from the Rutherford scattering experiment is that most of the mass of an atom is in the nucleus.

State **two** other conclusions from the experiment.

[2 marks]

1 _____

2 _____



Don't write

Which pair of particles can be produced in a single pair-production event?

[1 mark]

A electron and proton

☐

B electron and antineutrino

☐

C electron and positron

☐

D electron and neutron

☐

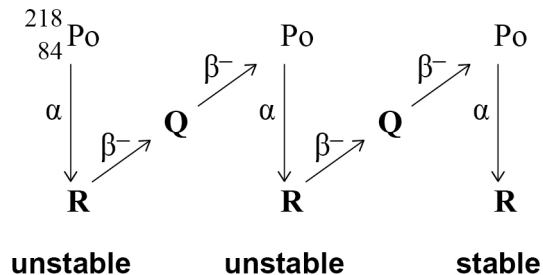
Turn over for the next question

Turn over ►



Don't write

A possible decay chain for the nuclide polonium-218 is shown.



Which row gives the nucleon number and proton number for the stable nuclide of element **R**?

[1 mark]

	Nucleon number	Proton number	
A	202	79	☐
B	212	76	☐
C	206	82	☐
D	212	82	☐

1 8 A particle has the following properties:

$$\text{charge} = +1.6 \times 10^{-19} \text{ C}$$

$$\text{mass} = 9.11 \times 10^{-31} \text{ kg}$$

What is the particle?

[1 mark]

A electron ☐

B neutrino ☐

C positron ☐

D proton ☐

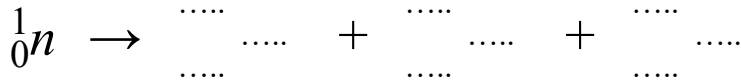


Don't write

0	3
---	---

Complete the equation for the decay of a free neutron.

[2 marks]



2



Don't write

Outline the Rutherford scattering experiment.

In your answer you should:

- draw a labelled diagram of the apparatus used
- describe the observations made
- explain what Rutherford deduced about the structure of the atom from these observations.

[6 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Turn over for the next question



Don't write

1	5
---	---

What is the order of magnitude of the ratio $\frac{\text{proton rest mass}}{\text{electron rest mass}}$?

[1 mark]

A -3

☐

B 1

☐

C 3

☐

D 1800

☐

Turn over for the next question

Turn over ►



Don't write